MFG-03: Carrington Storm Motors — Cost Reduction Pathway

Document ID: CSM/MFG/03 **Version:** 1.0 — Investor-Grade **Classification:** Confidential **Heuristic Foundation:** Accountant (learning curve economics), Chester (unit economics), El Segundo (cost trajectory) **Agents Consulted:** Manufacturing domain agent, Accountant, Chester **Date:** July 2026

1.0 EXECUTIVE SUMMARY

Carrington Storm Motors' path to profitability depends on aggressive manufacturing cost reduction from pilot-scale to volume production. The initial units — manufactured at pilot scale with manual processes, unoptimized supply chains, and low volumes — will cost 5-10x the target unit cost. Through learning curves, automation, process optimization, material substitution, and supplier leverage, CSM will reduce costs by 15-25% per doubling of cumulative volume until approaching the asymptotic minimum cost.

This document charts the cost reduction roadmap for each major product line, identifies the specific levers (automation, process optimization, material efficiency, supplier negotiation), and establishes cost targets at each volume milestone. The target: pilot-to-volume cost reduction of 80-90%, achieving competitive pricing while maintaining 40-55% gross margins.

As the Accountant Heuristic demands: "Every cost must earn its place in the product. A dollar of unnecessary cost is a dollar of grid protection not sold."

2.0 LEARNING CURVE ANALYSIS

2.1 The Manufacturing Learning Curve

Empirically, manufacturing costs follow a power law: the cost per unit decreases by a fixed percentage (the "learning rate") every time cumulative production doubles.

Industry	Typical Learning Rate (per doubling)
Aerospace	10-15%
Automotive	15-20%
Electronics	20-30%
Chemical processing	10-20%
Advanced materials (composites, ceramics)	15-25%
CSM Target (MXene-based products)	**20-25%**

CSM's learning rate target of 20-25% is aggressive but achievable given: - High initial manual labor content (large automation opportunity) - Novel processes with significant optimization potential - Low initial yields (yield improvement is the fastest cost reduction lever) - Volume-driven supplier cost reductions

2.2 Aegis Home — Cost Learning Curve (Illustrative)

Cumulative Units	Unit Cost (Relative to Target)	Primary Cost Reduction Levers
10 (pilot build)	500-1,000% of target	Manual assembly; pilot-scale material pricing; low yields (~50-70%)
100	250-500%	Process refinement; yield improvement (~70-85%); initial automation
1,000	125-250%	Semi-automated assembly; volume material pricing; yield >90%
10,000	110-150%	Automated production line; optimized supply chain; yield >95%

Cumulative Units	Unit Cost (Relative to Target)	Primary Cost Reduction Levers
100,000	100-120%	Target cost approaching; continuous improvement for final 10-20%
1,000,000	85-100%	Below-target cost; mature manufacturing

2.3 Charlemagne Fleet — Cost Learning Curve

Cumulative Substation Kits	Unit Cost (Relative to Target)	Notes
5	400-700%	First installations; custom engineering per site
50	200-350%	Standardized components; installation learning
500	120-180%	Volume manufacturing; installation crew proficiency
5,000	100-120%	Mature product; optimized logistics
50,000	85-100%	Commodity pricing on components

3.0 AUTOMATION OPPORTUNITIES

3.1 Automation Investment and Payback

Automation Project	Investment $(M) \vert Annual Labor Savings (K)$ Payback Period		Implementation (Phase)	
MXene film roll-to-roll line (full automation)	3-8	500-1,500	2-8 years	Phase 2
BFRP pultrusion automation (creel to cutoff)	2-5	300-800	2-7 years	Phase 2
Robotic assembly (Aegis Home)	3-8	500-2,000	2-8 years	Phase 2-3
Automated testing (in-line)	2-5	300-800	2-7 years	Phase 2
Automated warehousing (ASRS)	2-5	200-500	4-10 years	Phase 2-3
Packaging automation	1-3	100-300	3-10 years	Phase 2

3.2 Automation Sequencing

Automation is sequenced to maximize return on scarce capital: 1. **Highest manual labor content first:** Assembly, testing, packaging 2. **Quality-critical processes:** MXene film formation (consistency), testing (accuracy) 3. **Hazardous processes:** Chemical handling (safety + consistency) 4. **Lower priority:** Material handling, warehousing (Phase 3)

4.0 PROCESS OPTIMIZATION

4.1 Yield Improvement

Yield (percentage of produced units passing final inspection) is the single largest cost lever in early manufacturing:

Process	Phase 1 Yield (Est.)	Phase 2 Yield (Target)	Phase 3 Yield (Target)	Cost Impact
MXene synthesis (batch success rate)	60-80%	>90%	>98%	1.25-1.67x cost multiplier at Phase 1
MXene film formation (thickness uniformity)	50-70%	>85%	>95%	1.43-2.0x
BFRP pultrusion (profile quality)	70-85%	>92%	>97%	1.18-1.43x
Ceramic sintering (density/cracking)	65-80%	>90%	>96%	1.25-1.54x
Final assembly (functional test pass)	75-90%	>95%	>99%	1.11-1.33x
Overall First-Pass Yield (FPY)	**20-35%**	**>65%**	**>88%**	**2.86-5.0x** cost impact at Phase 1

4.2 Cycle Time Reduction

Process	Phase 1 Cycle Time	Phase 2 Target	Phase 3 Target
MXene etching + delamination	48-96 hours (batch)	24-48 hours	12-24 hours (continuous)
MXene film drying	24-48 hours (ambient)	4-8 hours (heated)	1-2 hours (continuous)
BFRP pultrusion (per meter)	0.5 m/min	1-2 m/min	3-5 m/min
Ceramic sintering	12-24 hours	6-12 hours	4-8 hours (continuous kiln)
Geopolymer curing	24-48 hours (ambient)	8-24 hours (heated)	4-8 hours
Final assembly (Aegis Home)	4-8 hours (manual)	1-2 hours	15-30 minutes

---

5.0 MATERIAL COST REDUCTION

5.1 Raw Material Cost Trajectory

Material	Phase 1 Unit Cost $(/kg) \vert Phase2Target(/$kg)	Phase 3 Target ($/kg)	Reduction Lever	
MAX phase (Ti3AlC2)	500-2,000	200-500	50-150	Volume; in-house synthesis
LiF	50-100	30-60	20-40	Volume purchasing
Basalt fiber roving	5-10	3-6	2-4	Volume; supplier competition
CNTs (MWCNT)	100-500	50-150	20-80	Volume; multi-supplier
Epoxy resin	5-10	3-6	2-4	Volume; commodity
BaTiO3 powder	20-50	10-25	5-15	Volume; alternative suppliers
Metakaolin	0.50-1.00	0.30-0.60	0.20-0.40	Regional sourcing
Alkali activator (geopolymer)	0.50-1.50	0.30-0.80	0.20-0.50	Volume; regional

5.2 Material Substitution Opportunities

Current Material	Substitution Candidate	Status	Cost Reduction Potential
MAX phase (Ti3AlC2)	Alternative MAX phases (V2AlC, Nb2AlC) for specific applications	Research	20-30% (vanadium vs. titanium)
MXene (high-conductivity grades)	Lower-conductivity MXene for non-critical shielding	Research	15-25% (less stringent processing)
Epoxy resin (thermoset)	Polypropylene (thermoplastic) for non-structural BFRP	Development	20-40% (cheaper resin + recyclability)
BaTiO3 ceramic	SrTiO3 or composite ceramics for lower-permittivity applications	Research	10-30%
CNT (high-purity)	Carbon black or graphene nanoplatelets for lower-conductivity applications	Development	50-80%
Geopolymer (metakaolin-based)	Fly-ash-based geopolymer (where fly ash locally available)	Implementation	20-40%

6.0 DESIGN FOR MANUFACTURABILITY (DFM)

6.1 DFM Principles Applied to CSM Products

DFM Principle	Aegis Home Example	Cost Impact
Part count reduction	Integrate MXene film + polymer substrate into single "dielectric panel"	20-30% fewer parts → less assembly labor
Common parts across products	Standardized MXene panels used in Aegis Home, Aegis Business, Charlemagne	Volume leverage on common components

DFM Principle	Aegis Home Example	Cost Impact
Modular design	Plug-and-play modules replaceable without full system disassembly	Reduced warranty/repair cost; field serviceable
Mistake-proofing (poka-yoke)	Keyed connectors that cannot be mated incorrectly	Reduces assembly errors; improves yield
Design for automated assembly	Flat, panel-based design suitable for robotic pick-and-place	Enables automation without redesign
Minimize fasteners	Snap-fit and adhesive bonding where structural requirements allow	Faster assembly; fewer part numbers
Standardized interfaces	Common connector types across all product lines	Supplier consolidation; inventory reduction

---

7.0 SUPPLIER NEGOTIATION LEVERAGE

7.1 Volume-Driven Cost Reduction

Volume Tier	Typical Discount from List Price	CSM Planned Volume
Pilot (<$100K annual spend/supplier)	0-10%	Phase 1
Low volume ($100K-$1M)	10-20%	Phase 1-2
Medium volume ($1-10M)	20-35%	Phase 2
High volume ($10-100M+)	30-50%	Phase 3
Strategic partnership ($100M+)	35-60% + co-investment	Phase 3

7.2 Negotiation Levers

Lever	Description
Long-term contracts (3-5 years)	Supplier trades price for volume certainty
Take-or-pay commitments	CSM guarantees minimum purchase → lower unit price
Co-development partnerships	Supplier invests in R&D for CSM-specific materials → lower price + exclusive access
Multi-year pricing agreements	Price escalation tied to published indices (not supplier discretion)
Competitive bidding	Regular re-bidding (every 2-3 years) to maintain price pressure
Supplier financing	CSM helps supplier finance capacity expansion → lower price in exchange

---

8.0 ENERGY COST OPTIMIZATION

8.1 Energy Consumption Profile

Process	Energy Intensity	% of Total Manufacturing Cost (Phase 2)
MAX phase synthesis (1,400-1,600°C sintering)	VERY HIGH	10-20%
Ceramic sintering (1,200-1,500°C)	VERY HIGH	10-15%
MXene drying (heated)	HIGH	5-10%
BFRP pultrusion (curing ovens)	MODERATE-HIGH	5-10%
Geopolymer curing (heated)	MODERATE	2-5%
Facility HVAC/lighting	MODERATE	3-5%
Other	LOW	5-10%

8.2 Energy Cost Reduction Strategies

Strategy	Savings Potential	Implementation
Renewable energy PPA (Power Purchase Agreement)	20-40% vs. grid electricity	Phase 2-3
Waste heat recovery (sintering → drying)	15-25% reduction in drying energy	Phase 2
On-site solar + battery storage	10-30% of facility electricity	Phase 2-3
High-efficiency furnaces/kilns	10-20%	Phase 1 (specify in equipment procurement)
Demand response participation	Revenue from grid operator for load shedding	Phase 2-3
Location selection (low electricity cost regions)	20-50% vs. high-cost regions	Phase 1 (facility siting decision)

9.0 TARGET COST TRAJECTORY BY PRODUCT LINE

9.1 Aegis Home (Standard)

Volume Tier	Unit Manufacturing Cost	Target Selling Price	Gross Margin
Pilot (100 units)	$15,000-25,000	$2,500-5,000 (subsidized)	Negative (investment)
Initial volume (1,000)	$5,000-8,000	$3,500-5,000	15-30%
Mid volume (10,000)	$2,000-3,500	$2,500-5,000	28-50%
High volume (100,000)	$1,200-2,000	$2,000-4,000	40-55%
Mass market (1,000,000+)	$800-1,200	$1,500-3,000	45-60%

9.2 Charlemagne Fleet (Per Substation Kit)

Volume Tier	Unit Manufacturing Cost	Target Selling Price	Gross Margin
Pilot (5 installations)	$3-5M	$1.5-2.5M (cost-sharing with pilot utility)	Negative
Early deployment (50)	$800K-1.5M	$1-2M	20-35%
Volume (500)	$400-700K	$800K-1.5M	35-50%
High volume (5,000)	$250-450K	$500K-1M	40-55%

10.0 BREAKEVEN VOLUME ANALYSIS

10.1 Facility-Level Breakeven

Facility	Annual Fixed Costs ($M)	Contribution Margin per Unit	Breakeven Units/Year
Pilot facility (Phase 1)	3-6	Negative initial contribution	No standalone breakeven (investment phase)
Volume facility (Phase 2)	15-35	$500-3,000 (varies by product mix)	10,000-70,000 units (product mix dependent)
Global network (Phase 3)	80-200	$500-3,000	60,000-400,000 units

10.2 Corporate-Level Breakeven

Year	Revenue $(M)\vert TotalCosts$(M)	Operating Income	Notes	
Year 1	0	6-12	($6-12M)	R&D phase
Year 2	2-5	15-25	($13-20M)	First pilot revenue
Year 3	10-25	25-45	($15-20M)	Commercial launch
Year 4	40-80	50-90	($10M)	Approaching breakeven
Year 5	100-200	90-180	Breakeven to $20M	Target: profitability
Year 6	200-400	160-340	$20-60M	Sustained profitability
Year 7	500-800	375-640	$50-160M	IPO-ready

11.0 AGENT VOICES

The Accountant (Cost Trajectory): “The learning curve saves every manufacturing company — but only if management has the discipline to track costs obsessively, eliminate waste relentlessly, and invest in automation at the right moment (not too early, when processes are changing, and not too late, when competitors have already locked in cost advantage). The Accountant Heuristic says: know your costs better than your competitors know theirs. Every unit produced must be cost-tracked. Every process must have a cost model. Every investment must have a quantified payback.”

Manufacturing Domain Agent: “The MXene yield at pilot scale will be terrible. 50-70% first-pass yield means 30-50% of everything we make goes into the scrap bin. That’s normal for novel materials at pilot scale. What matters is the slope of the yield improvement curve — how fast we learn and optimize. A 5% yield improvement per month compounds to 80% improvement in 36 months. The key metric is not the starting yield — it’s the rate of improvement.”

Chester (Unit Economics): “At the corporate level, breakeven in Year 5 is ambitious but achievable. It requires: (1) manufacturing cost reduction on plan, (2) insurance revenue stream contributing 10-20% of revenue by Year 4-5, (3) government contracts with favorable payment terms, and (4) disciplined spending on G&A and R&D as a percentage of revenue. The most important number: gross margin trajectory. If gross margins are expanding quarter-over-quarter, the breakeven math works. If they’re flat, something is broken.”

Document prepared for investor and operational planning. All cost estimates are forward-looking and subject to significant uncertainty. Actual costs will depend on manufacturing execution, supplier negotiations, automation success, and yield improvement rates. Learning curve projections are illustrative and not guaranteed.